

- RCNN**
- Uses selective search ~ 2000 regions
 - CNN runs separately on each region
 - SVM for classification.
 - Bounding Box regressor
 - slow (as CNN runs on all regions)

- Fast RCNN**
- CNN runs once on full image
 - Produces feature map.
 - Regions are mapped to feature map.
 - *selective search* - ROI pooling will convert variable size regions to fixed size feature vector.
 - after ROI $\rightarrow$ flatten features $\rightarrow$ fully connected layers. (Per region)
 - 2 o/p's - Branch 1 classification
Branch 2 Bounding Box regressor

- Faster RCNN**
- selective search removed.
 - RPN (region proposal n/w) = small neural n/w sliding over feature map.
 - RPN predicts objectness score & Bounding Box by sliding over feature map

- RPN predicts anchors per location
then assigns probability score to
each anchor then keeps top k
(NMS & thresholding)

- CNN



full image



feature map

RPN

anchor

obj score

BB coordinates

Top K

← RoI Pooling
and FC layers.

Mask RNN

- extends faster RNN

- RoI align (don't use
quantized stride)

- RoI Pool uses 2 if stride 2.42
(misalignment, data loss)

RoI align uses 2.42

(maintains spatial accuracy)

- adds Mask branch.

B1 - class

B2 - BB

B3 - Pixel-wise mask.